MISH 27
Construction Equipment Modifications and Fabrications

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Construction Equipment Modification & Fabrication

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MH CONSTRUCTION INDUSTRIAL SAFETY AND HEALTH PROGRAM

Title

CONSTRUCTION EQUIPMENT MODIFICATIONS AND FABRICATION

Number

Revision 1 Approved By Jeremy Thamert 03/01/2025 Name Date Effective Date 3/1/2025 Page 1 of 3

1.0 PURPOSE

To define the requirements and responsibilities for control of construction equipment modifications, fabrications, and special purpose use in order to maintain safe operating conditions. Further, to ensure compliance with American National Standards Institute (ANSI), Occupational Safety and Health Administration (OSHA), and Washington Industrial Safety and Health Act (WISH) requirements, and the specifications of the manufacturer or registered Professional Engineer

2.0 RESPONSIBILITY

2.1 The MH Construction Industrial Safety and Health Department (IS&H) is responsible for review and evaluation of proposed modifications. IS&H may require preparation of a formal Job Hazard Analysis (JHA) to include special operator training requirements.

2.2 The Key Supervisor shall be responsible for the use of equipment assigned to their areas. This responsibility includes the assignment of construction equipment to the crafts for use and for maintenance of that equipment

MH Construction Management shall be responsible for the design and/or review of modifications or fabrication of proposed construction equipment. This includes the preparation of design drawings, and alteration procedures, including the necessary nondestructive examinations as required.

3.0 DEFINITIONS

Construction Equipment: Equipment used in the construction industry. This includes, but is not limited to, the following items:

• Crane

• Hoists

• Light vehicles

• Trucks

• Earth-moving equipment

• Tools and machines

• Drilling Rigs

Equipment Modification: Alterations of construction equipment from the original configuration as received and/or designed by the equipment manufacturer.

Equipment Fabrication: Construction equipment designed and/or built on the jobsite. Equipment that is not manufactured but originated on the construction site may be original in design or have similarities to manufactured equipment.

Special Purpose or Nonstandard Use: Use of construction equipment for operations not originally considered by the equipment manufacturers.

4.0 GENERAL REQUIREMENTS

4.1 MH Construction Project Management and MH Construction IS&H shall be notified prior to modifications or fabrications of equipment.

4.2 A thorough review of the equipment shall be performed by MH Construction Project Management and IS&H.

5.0 PROCEDURE

The process required to modify construction equipment is shown in attachment MISH 27-A.

5.1 Any request for modifications, fabrication or nonstandard use of construction equipment shall be made to MH Construction Project Management. All requests shall contain the following information as a minimum:

5.1.1 A description of the equipment, including information such as manufacturer's model number, capacity, limitations on use, etc.

5.1.2 The reason for the request, including any information on availability of alternate equipment, schedule impact, cost effect, etc.

5.1.3 A description of the modifications, fabrications and/or special purpose use of construction equipment. Drawings or sketches of original equipment or configuration and the proposed alteration/fabrication shall be made available to MH Construction Project Management.

5.2 MH Construction Project Management shall review the request for modifications, fabrications, and/or nonstandard use in regard to the overall project need. This review includes consideration of the following:

5.2.1 Use of alternate equipment that may be available.

5.2.2 Direct purchase of special equipment to meet the desired needs.

5.2.3 Manufacturer's specifications.

5.2.4 Safeness of the request.

IS&H will review the request for modification, fabrication and/or nonstandard use for compliance with the appropriate safety standards, and will consider special safety requirements imposed by IS&H, and either accepts or rejects the request. All requests for modification/fabrication shall be returned to MH Construction Project Management. Request, which have been rejected, shall include an explanation.

5.3 MH Construction Project Management shall review the design for the modification or fabrication based on the information received. An engineer will conduct this review. The design includes a review of the structural integrity of the modification/fabrication and any other review deemed appropriate.

5.4 Approved request shall be forwarded to IS&H for a final safety check along with any design drawings and procedure. A JHA may be required at this point. Upon final approval, the request is released to the Contractor for implementation and use.

5.5 Upon final approval, the MH Construction shall fabricate or modify equipment according to design.

6.0 RECORDS

Document Record Submittal Responsibility Record Retention Responsibility Requests, Drawings, Approvals, etc.

Contractor Project Management

7.0 REFERENCES

29 CRF 1910 Occupational Safety and Health Standards (OSHA) 29 CRF 1926 Safety and Health Regulations for Construction (OSHA) WAC 296-155 Safety Standards for Construction Work (WISHA)

8.0 ATTACHMENTS N/A

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